Supplementary Information
Road-supported effective constitutive coupling and anisotropic routing responses in urban mobility

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Supplementary Methods

Notes S1–S5 specify the three-city data, fields, road registration, estimators, scores, uncertainty, joint spectra, coherent response, anisotropy, matched-capacity media and comparison protocols. Figures are numbered by first appearance and interpreted in the corresponding results Notes.

Note S1 Data versions, observation regimes and frozen splits

The focal records are Porto taxi trajectories from the ECML/PKDD 2015 challenge [1], San Francisco Cabspotting trajectories (CRAWDAD `epfl/mobility`, version 2009-02-24) [2], and an archived 2014 DiDi ride-hailing sample from Chengdu [3], with regularly spaced 30-second fixes in the aligned protocol. Table S1 separates releases, observation regimes and frozen date protocols. Errors are never pooled across cities or observation regimes.

Table S1: Three-city observation regimes and frozen date splits. TRAIN/VAL/TEST counts are dates. San Francisco has five blocked folds, each 9/3/3. Chengdu's three protocols are related but not interchangeable.

Object	Data version or release	Regime and frozen protocol	Split	Lattice	Hours
Porto	ECML/PKDD 2015	Taxi; 252 eligible dates; chronological	151/50/51	32×32	00–23
San Francisco	Cabspotting v. 2009-02-24	Taxi; 15 dates; five blocked folds	9/3/3 per fold	32×88	00–23
Chengdu: P/D, spectra, anisotropy	Archived DiDi 2014 sample	Continuous-observation subset; 14 dates	8/3/3	60×60	06–23
Chengdu: structural null	Same aligned grid	All 19 valid dates; full aligned observation pool, not a P/D fit	–	60×60	aligned pool
Chengdu: matched capacity	Same archived corpus	Chronological 18-date protocol; four TEST dates 26–29 Aug. 2014 use alternate-hour delivery, nine recorded hours/date	10/4/4	60×60	alternating subset of 06–23

Chengdu's 14-date subset has three TEST dates in the continuous-observation regime and supplies the P/D values in Table S3. Its 18-date chronological split instead supplies the matched-capacity experiment: the four held-out dates fall wholly in the alternate-hour delivery phase. The 19-date pool is used only to compare representation subspaces. These different date counts and observation regimes must not be merged. The design cannot separate the influence of date count, delivery phase and channel on the matched-capacity effect; its four-date interval is only a descriptive small-sample summary.

The nominal cell size is 250 m on the focal aligned grids. Successive trajectory segments are walked through aligned cells and accumulated as signed interior-face crossings in 60-minute windows. No map matching is applied. The archived analysis footprint is the union of cells with non-zero density over all dates, $(N > 0).\mathrm{any}(0)$; it is therefore transductive. It is used only to restrict analysis

and display cells and is not an input to D, U_R or K_R, which depend on the grid, study domain and road registration. A TRAIN-only footprint is a strict subset (987 versus 994 cells in Porto, 2345 versus 2532 in one San Francisco fold), but the core operator does not change because the mask does not enter its construction. The flux-derived source is the divergence of the observed crossing field, not an independently measured trip-production or OD-demand field. Unrecorded raw-data schema or preprocessing fields are not reconstructed from their names or presumed to be available.

Attribution, licensing and release checks. Road geometry derives from the drivable OpenStreetMap layer [4]; maps and derived road products retain the attribution © *OpenStreetMap contributors*. The current archive does not provide sufficient per-extract metadata to verify the OSM snapshot dates/versions, the licence notices accompanying every derived artifact, or all downstream redistribution obligations. Those items must be checked against the source extracts before public release; no city-specific version or licence clearance is invented here. The exact public release/version and redistribution terms of the archived Chengdu sample likewise remain unverified. Public data availability does not imply that individual-level trajectories are included in the export.

Note S2 Fields, road registration, estimation and uncertainty

Discrete fields and conservation. Let $F_t \in \mathbb{R}^{n_f}$ be the observed signed crossing field on interior faces; a prediction is distinguished by a hat. The cell–face incidence matrix $D \in \mathbb{R}^{n_c \times n_f}$ maps face flux to net cell outflow under the frozen orientation. Thus

$$x_t = DF_t, \qquad P_0 = I - D^\dagger D, \qquad F_t^\circ = P_0 F_t, \qquad DF_t^\circ = 0. \tag{S1}$$

The observed source–sink balance and divergence-free routing field give the spatially orthogonal decomposition $F_t = D^\dagger x_t + F_t^\circ$. Orthogonality at an instant does not imply statistical independence across observations. The minimum-norm lift uses a gauge-pinned Laplacian solve rather than an explicitly formed Moore–Penrose inverse; on the zero-sum compatible source space the two constructions coincide. Conservation is imposed by construction or projection, not learned.

Road registration. The registered road graph uses road nodes and registered road segments as edges; it is not trimmed to the mobility footprint. If B_R is its node–edge incidence, A_R lifts signed road-edge flows to interior faces, and Π aggregates cells to nodes, registration satisfies $DA_R = \Pi B_R$ in integer arithmetic. For a cycle basis Z_R with $B_R Z_R = 0$,

$$U_R = \mathrm{orth}(A_R Z_R), \qquad U_R^\top U_R = I, \qquad DU_R = 0. \tag{S2}$$

The signed entries of A_R count open-segment crossings in the frozen face orientation. Geometry, study domain and road registration determine the basis; the analysis footprint mask, mobility values, dates and folds do not. This independence is conditional on the registered spatial conventions. The observed road-coordinate state is $y_t = U_R^\top F_t^\circ$. The physical lift $C_R = U_R K_R P_x$, with $P_x = DD^\dagger$, is used for the local response tensors in Note S4.

Response estimation. For any source or target v_t, TRAIN phase means and departures are

$$\bar{v}_\phi = \frac{1}{n_{\mathrm{tr},\phi}} \sum_{t \in \mathrm{TRAIN}:\phi(t)=\phi} v_t, \qquad \delta v_t = v_t - \bar{v}_{\phi(t)}. \tag{S3}$$

Stack $\delta x_t^\top$ and $\delta y_t^\top$ as rows of X and Z. The road-coordinate ridge response is

$$K_R^\star = \arg\min_K \{\|Z - XK^\top\|_F^2 + \lambda\|K\|_F^2\} = Z^\top X(X^\top X + \lambda I)^{-1}, \tag{S4}$$

with $\lambda = \kappa s_{\max}(XX^\top)$ and $\kappa \in \{10^{-10}, 10^{-8}, 10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100\}$. Validation selects κ in the scoring frame of each comparison arm. The structural-characterization estimators use $\kappa = 0.01$ in Porto, 0.1 in San Francisco, 0.1 in Chengdu's D frame, and 0.001 for Chengdu's P-frame sensitivity. Each city is fitted separately; these settings are not a cross-city transfer rule.

P, D and CLIM scores. Let $T_t = U_R^\top F_t^\circ$ on ROAD support and $T_t = F_t^\circ$ on FACE support. ROAD scoring projects full-face predictions into road coordinates; FACE scoring lifts road predictions to the complete routing field, so their denominators differ. Both fitted maps use TRAIN departures. P selects on raw VAL inputs and targets and predicts $B_{\mathrm{P}} x_t$ without a fitted intercept; D centres VAL/TEST with TRAIN phase means and predicts $B_{\mathrm{D}}\delta x_t$:

$$\begin{aligned} R^2_{\mathrm{P}} &= 1 - \frac{\sum_{t\in\mathrm{TEST}}\|T_t - B_{\mathrm{P}} x_t\|^2}{\sum_{t\in\mathrm{TEST}}\|T_t\|^2}, \\ R^2_{\mathrm{D}} &= 1 - \frac{\sum_{t\in\mathrm{TEST}}\|\delta T_t - B_{\mathrm{D}}\delta x_t\|^2}{\sum_{t\in\mathrm{TEST}}\|\delta T_t\|^2}. \end{aligned} \tag{S5}$$

Numerators and denominators are pooled over TEST observations and folds within each city. The denominator is target energy, not variance about a TEST mean. CLIM is the source-free prediction $\bar{T}_{\phi(t)}$ on the same support and samples. For the P-frame physical display in Figure S6, the observed field is $U_R T_t$ and the prediction is $U_R B_{\mathrm{P}} x_t$, using raw TEST inputs and targets without adding a phase climatology or substituting a D departure.

Full-face ridge fits F_t° and projects its predictions with P_0. POD regression estimates its output basis from TRAIN departures only and selects rank and regularization on VAL in the common FACE metric. No TEST observation selects a basis, rank or regularization. These linear controls and the matched-capacity protocol in Note S4 are distinct experiments.

Nulls and uncertainty. The rank-matched structural null fixes the ambient space and basis rank but not the subspace. Let K_D be an orthonormal basis of $\ker D$. Each draw uses $G_{ij} \overset{\mathrm{iid}}{\sim} N(0,1)$, $Q = \mathrm{orth}(G)$ and $U_b = K_D Q$ at $\mathrm{rank}(U_R)$. The statistic is

$$E(U) = 1 - \frac{\sum_t \|U^\top F_t^\circ\|^2}{\sum_t \|F_t^\circ\|^2}, \qquad \Delta E_b = E(U_b) - E(U_R), \tag{S6}$$

evaluated on each city's full aligned observation pool without a source-response fit. It tests representation specificity, not source-based prediction.

The primary three-city D scores in Table S3 have paired-date bootstrap intervals and source–routing pairing-null results, reported in Note S6. Dates are resampled with replacement within each fold; faces are not resampled. The pairing null permutes complete date blocks within TRAIN, VAL and TEST separately and refits the operator in every draw. Chengdu has only three dates, so its interval and null are descriptive. The neural paired-date contrast uses 9,999 bootstrap resamples, seed 20261003, dates resampled within folds and Holm adjustment within city/support. The matched-capacity implementation uses 10,000 multinomial paired-date resamples with seed 23 and summarizes the date-mean contrast. Chengdu's matched-capacity interval resamples only four alternate-hour TEST dates and is a small-sample descriptive interval.

Note S3 Joint spectra and coherent response

Joint temporal–spatial spectrum. For raw field $v_{d,j}$ on a TRAIN-date block of length N, a unitary one-sided discrete Fourier transform gives

$$\begin{aligned} Z_{d,m} &= N^{-1/2} \sum_{j=0}^{N-1} v_{d,j} e^{-2\pi i m j/N}, \qquad f_m = m/N, \\ P_m &= w_m \langle \|Z_{d,m}\|^2 \rangle_d, \qquad J_{m,k} = \frac{P_m}{\sum_l P_l} \frac{s_{m,k}^2}{\sum_a s_{m,a}^2}. \end{aligned} \tag{S7}$$

Here $s_{m,k}$ are singular values of the date-by-space coefficient matrix, $w_0 = w_{N/2} = 1$, and interior weights are two. The transform retains $m = 0$, so the spectra include each date's phase mean. Porto and San Francisco use $N = 24$; Chengdu uses $N = 18$ for 06–23 h without night imputation or zero filling. San Francisco averages P_m and within-frequency mode shares separately across TRAIN folds before multiplication. Normalization uses the full weighted spectrum before the eight-mode display cut in Figure S1. Spatial modes are frequency-specific singular modes, not geographical Fourier wavelengths.

Chengdu has ten single-sided bins $m = 0, \ldots, 9$ at $m/18$ cycles per observed hour. The periodic extension joins the ends of the observed block; it does not infer the unobserved night. A 24-hour sinusoid is off this grid, so the first nonzero bin is not identified as the once-daily harmonic. Zero padding to 24 samples would change the frequency grid and redistribute shares, not establish zero night-time flow.

Coherent TRAIN/TEST response. The frozen P-selected operator is applied without refitting a frequency response. Each date is transformed with the same unitary DFT support as the descriptive spectrum: 24 points in Porto and San Francisco and 18 points in Chengdu. Let $a_{d,m}$ and $b_{d,m}$ be observed and predicted routing coefficients on date d at single-sided bin m, and let $a_m^{(\mathcal{S})}$, $b_m^{(\mathcal{S})}$ be their averages over split $\mathcal{S} \in \{\text{TRAIN}, \text{TEST}\}$. The phase-insensitive magnitude and norm ratio are

$$\gamma_m^{(\mathcal{S})} = \frac{|a_m^{(\mathcal{S})*} b_m^{(\mathcal{S})}|}{\|a_m^{(\mathcal{S})}\| \|b_m^{(\mathcal{S})}\|}, \qquad \rho_m^{(\mathcal{S})} = \frac{\|b_m^{(\mathcal{S})}\|}{\|a_m^{(\mathcal{S})}\|}. \tag{S8}$$

The absolute inner product makes γ_m insensitive to a common phase rotation, so it cannot by itself validate a complex response. We therefore also report the per-date phase-sensitive score and signed alignment

$$R_m^2 = 1 - \frac{\sum_d \|a_{d,m} - b_{d,m}\|^2}{\sum_d \|a_{d,m}\|^2}, \qquad \widetilde{\gamma}_m = \frac{\sum_d \mathrm{Re}(a_{d,m}^* b_{d,m})}{\sqrt{\sum_d \|a_{d,m}\|^2} \sqrt{\sum_d \|b_{d,m}\|^2}}. \tag{S9}$$

The signed alignment pools date-wise complex inner products and energies across TEST dates; it is not a cosine of date-averaged coefficients. TRAIN values describe the fit sample; TEST values are held-out. San Francisco summaries are fold means and ranges. Chengdu evaluates all ten bins $m = 0, \ldots, 9$. Figure S3 displays the TRAIN/TEST magnitudes, while the phase-sensitive TEST scores are shown in main Fig. 3. Across Porto bins, R_m^2 falls from 0.89 at $m = 0$ to 0.20 at $m = 12$ while $|\gamma_m|$ remains 0.46 and pooled signed alignment spans 0.46–0.96; San Francisco has negative R_m^2 at four bins, and Chengdu at one. The phase-insensitive alignment therefore overstates the agreement between predicted and observed complex responses.

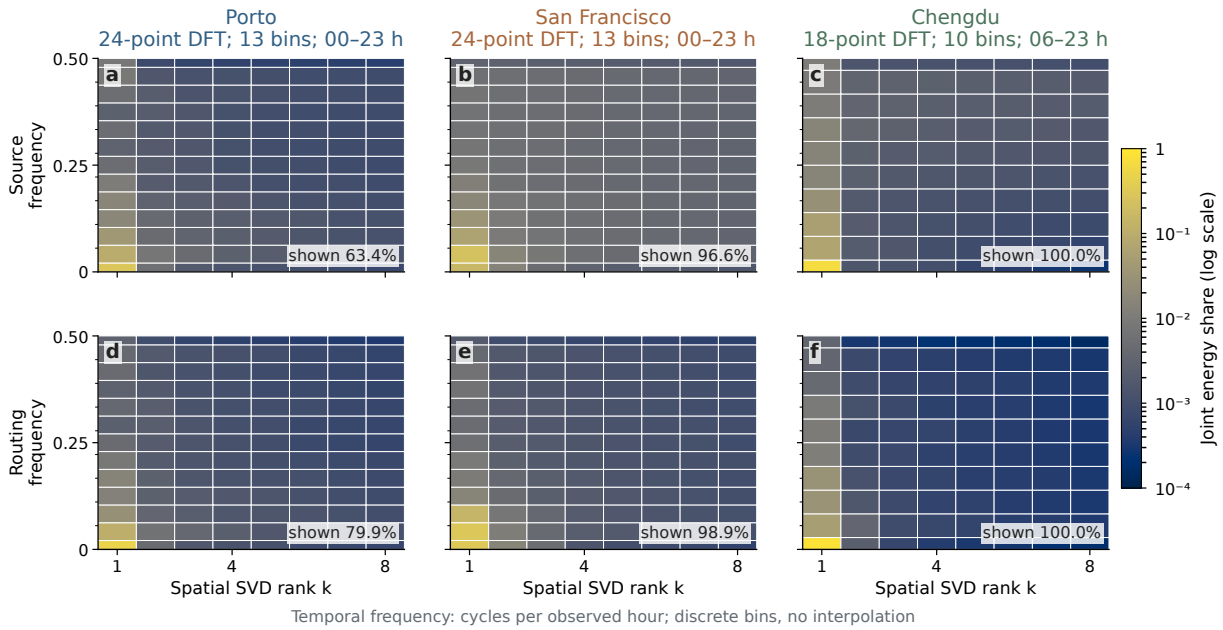


Figure S1: Three-city two-dimensional joint spectra. Joint energy share over temporal frequency and frequency-specific spatial mode rank for source and full-face routing fields in Porto, San Francisco and Chengdu. The eight-mode display cut is applied after full-spectrum normalization. Porto and San Francisco use 24-point DFTs; Chengdu uses an 18-point observed-block DFT with no night imputation. San Francisco averages frequency energies and within-frequency mode shares separately across TRAIN folds. Spatial rank is not geographical wavelength.

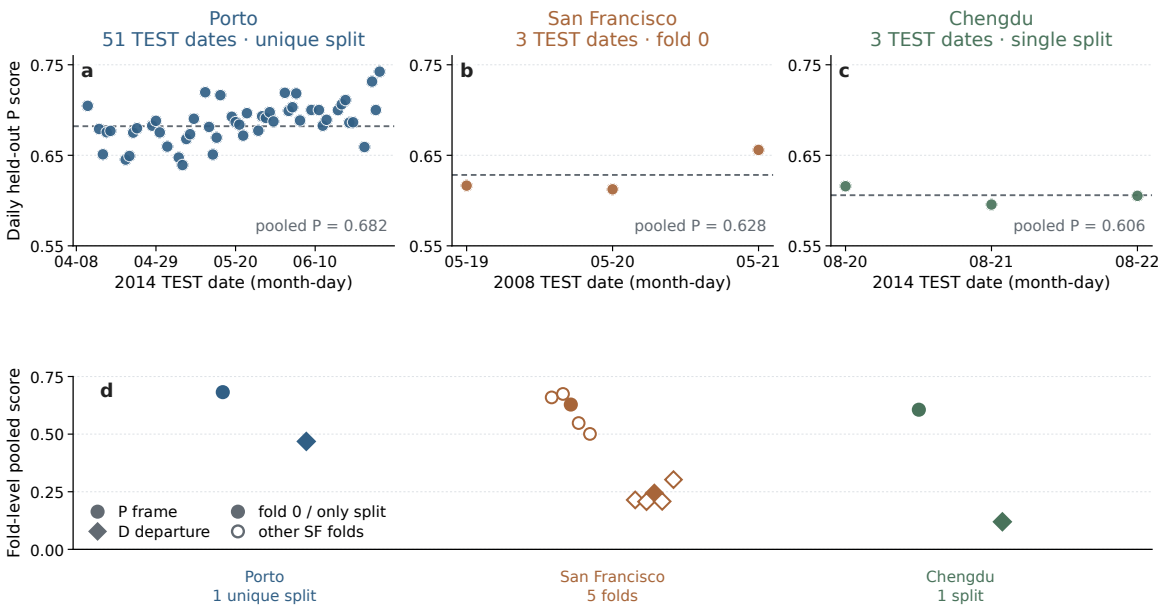


Figure S2: Three-city held-out daily P/D scores. Date-specific P and D values for Porto, San Francisco and Chengdu under the frozen splits in Table S1. San Francisco retains its five-fold structure; Chengdu has three TEST dates in the continuous-observation protocol. Daily values complement the pooled scores in Table S3; their spread is not a paired-date confidence interval.

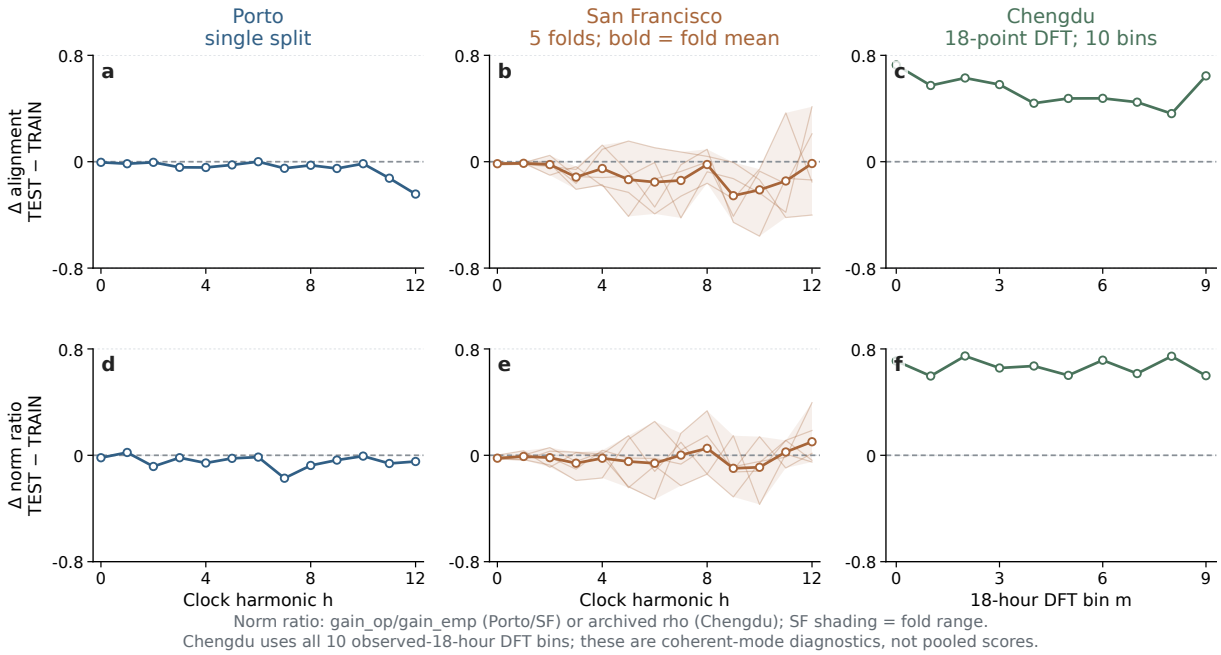


Figure S3: Three-city coherent response on TRAIN and TEST. Direction alignment γ_m and predicted-to-observed norm ratio ρ_m for Porto, San Francisco and Chengdu under the frozen P-selected operator. TRAIN values describe the fit sample; TEST values are held-out. San Francisco shows fold means and ranges. Each date uses the same unitary DFT support as the descriptive spectrum; Chengdu evaluates all ten single-sided bins of its 18-point transform. Alignment is insensitive to a common phase rotation.

Note S4 Anisotropy and matched-capacity media

Local tensors and axial orientation. For each regular interior cell, L_c maps signed face flux to two physical components. Define

$$\begin{aligned}\Sigma_C(c) &= L_c C_R C_R^\top L_c^\top,\\ \Sigma_U(c) &= L_c U_R U_R^\top L_c^\top,\\ A(c) &= \begin{cases}(\lambda_1(c)-\lambda_2(c))/(\lambda_1(c)+\lambda_2(c)), & \lambda_1(c)+\lambda_2(c)>0,\\ 0, & \lambda_1(c)+\lambda_2(c)=0.\end{cases}\end{aligned} \tag{S10}$$

Here $\lambda_1 \geq \lambda_2$ are eigenvalues of the relevant tensor. The principal eigenvector defines an axis θ modulo π, not a directed arrow. San Francisco uses per-cell mean anisotropy and a nematic mean of $e^{2i\theta}$ over five folds. Directional coherence is

$$S_{\mathrm{nem}} = \frac{\left|\sum_c w_c A_C(c) e^{2i\theta_C(c)}\right|}{\sum_c w_c A_C(c)}, \tag{S11}$$

with $w_c = 1$ or $w_c = \operatorname{tr}\Sigma_C(c)$.

Structural-axis agreement is the joint-anisotropy estimand

$$G_\theta = \frac{\sum_c A_C(c) A_U(c) \cos[2(\theta_C(c)-\theta_U(c))]}{\sum_c A_C(c) A_U(c)}, \qquad A_C(c)A_U(c) > 0, \tag{S12}$$

on regular interior cells for which both anisotropies are nonzero. Because Σ_C uses a unit input metric, we also evaluate the source-covariance-weighted response tensor

$$\Sigma_{\mathrm{response}}(c) = L_c C_R \Sigma_{xx} C_R^\top L_c^\top, \tag{S13}$$

with Σ_{xx} the TRAIN covariance of source departures. Median anisotropies under this metric are 0.766, 0.926 and 0.765 in Porto, San Francisco and Chengdu, compared with 0.635, 0.651 and 0.705 under the unit metric. Strong local anisotropy persists under source-covariance weighting, indicating that directional selectivity is also pronounced under the realized excitation statistics. This comparison does not establish independence from every possible input metric.

Axis agreement is a separate question because C_R and U_R share the road subspace by construction. We write $K_R = Q\Lambda V^\top$ and replace Q by a Haar-random orthonormal frame Q_0 in the road output space, $K_0 = Q_0\Lambda V^\top$, preserving the spectrum and input directions. Recomputing G_θ for 999 draws gives null minima of 0.967, 0.902 and 0.803 and medians of 0.973, 0.911 and 0.808 in Porto, San Francisco and Chengdu. The archived estimates (five-fold consensus for San Francisco), 0.79, 0.79 and 0.76, lie below every null draw, so no observed value exceeds the null (one-sided empirical $p = 1.0$ in all three cities). The archived tensors in `full_operator_spatial_anisotropy/results/fo4_fig5_data.npz` and `reports/threecity_figure5_revision/results/chengdu_anisotropy.npz` are the single primary source for these estimates and for Fig. 4a–c. Because the rotation requires K_R, which is not stored with those tensors, the null is generated by a control refit at the archived κ and a single-fold reconstruction (San Francisco fold 0). This control reproduces the archived scale but is not used as an estimate; the reported conclusion uses only the archived estimates against the null quantiles, and the provenance audit records both the archived sources and the control's role. Because the control's own G_θ is at least as large as the archived estimate in every city, the

comparison is conservative for the reported conclusion. Axis agreement is not distinguishable from the spectrum-matched null and provides no independent evidence of a road-aligned response. We retain only the anisotropy magnitude as an operator property.

On the regular interior grids $L_c L_c^\top = 2I$, so metric whitening is a no-op and grid anisotropy vanishes there; anisotropic boundary-cell metrics are excluded. An orthogonal road-coordinate rotation leaves the physical tensor unchanged when basis and coefficients are transformed together. Under a rigid physical rotation Q,

$$\Sigma_C(c) \mapsto Q\Sigma_C(c)Q^\top, \qquad v_C(c) \mapsto Qv_C(c), \tag{S14}$$

while eigenvalues and scalar anisotropy are invariant. The axial vector is covariant, not numerically unchanged in fixed physical coordinates.

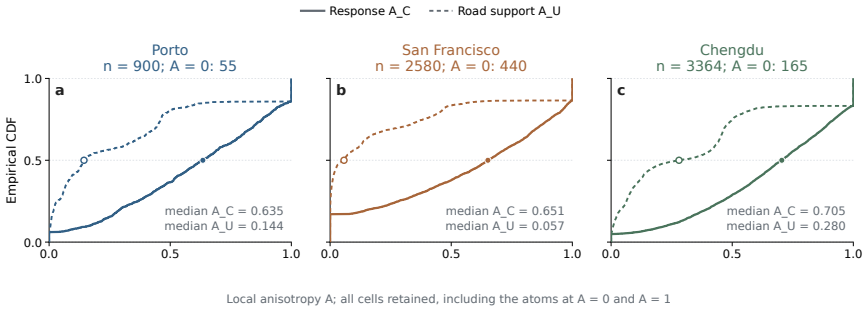


Figure S4: Three-city local anisotropy distributions. Empirical cumulative distribution of $A = (\lambda_1 - \lambda_2)/(\lambda_1 + \lambda_2)$, with $A = 0$ assigned to a zero tensor, over regular interior cells for road support A_U (dashed) and fitted response A_C (solid); circles mark medians. Both curves use the same cells and per-cell definition. Endpoint steps are real atoms: cells without directional energy enter at $A = 0$ (5–17%, identical cell sets for A_U and A_C), and one-dimensional tensors enter at $A = 1$ (6–16%). They are not normalization artefacts.

Matched-capacity media. The compared media are a road medium from an external GIS raster, a mobility-inferred medium estimated from TRAIN mobility observations, and a permuted mobility medium. At capacity R, each per-face source pattern is averaged over R spatial blocks and passed through the verified mean-1/floor normalization

$$w_{S,f}^{(R)} = \max\left(\frac{\overline{W}_{S,f}^{(R)}}{\langle \overline{W}_S^{(R)} \rangle_{\text{interior faces}}}, 0.02 \right), \tag{S15}$$

before the archived minimum-energy field lift. Thus R is the nominal number of block values before normalization, not the effective degrees of freedom: mean-1 normalization removes at least one independent degree, and the floor can reduce it further. With $R = 1$ every medium is constant, and $R = n_{\text{int}}$ assigns one pre-normalization value per interior face. Permutation preserves the mobility medium's values, partition and nominal block count while permuting its spatial arrangement. Road and mobility media do not have identical information inputs; equal R matches capacity, not information.

The score is computed per held-out date against that date's own target mean and then averaged with equal date weight:

$$R^2_{\text{cond},d} = 1 - \frac{\sum_{t\in d} \|T_t - \widehat{T}_t\|^2}{\sum_{t\in d} \|T_t - \langle T \rangle_d\|^2}. \tag{S16}$$

It is an archived date-centred statistic on the raw routing target, not on D's TRAIN-phase departure target, and is distinct from P and D in Eq. S5. Figure S5 reports gains over the constant $R = 1$ medium.

The nested grids are Porto $R = (1, 4, 16, 64, 256, 1984)$ with block sides $(0, 16, 8, 4, 2, 1)$; San Francisco $R = (1, 3, 12, 44, 176, 704, 5512)$ with block sides $(0, 32, 16, 8, 4, 2, 1)$; and Chengdu $R = (1, 4, 16, 64, 225, 900, 7080)$ with block sides $(0, 32, 16, 8, 4, 2, 1)$. Side zero denotes the single constant block. The final identity partition gives $R = n_{\text{int}} = (1984, 5512, 7080)$. San Francisco uses its occupancy-defined channel and per-fold basis; Chengdu has only the archived all-motion channel and uses the frozen per-fold-basis rule on its chronological 10/4/4 split.

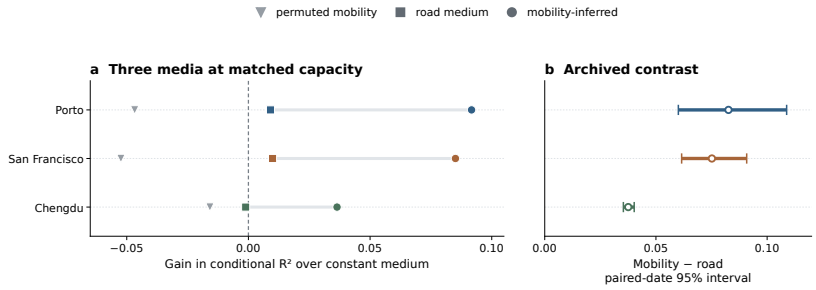


Figure S5: Three-city media at matched capacity. Gain in held-out conditional R^2 over the constant $R = 1$ medium for the permuted mobility medium, road medium and mobility-inferred medium at $R = n_{\text{int}}$ (1984, 5512 and 7080 nominal pre-normalization block values). The score is the equal-date mean of within-date conditional R^2, distinct from P and D. Constant-medium values are 0.613, 0.743 and −0.030. Permutation preserves values, partition and nominal block count. Chengdu resamples four alternate-hour dates on a single all-motion channel; its points are small-sample summaries.

The matched-capacity modules `domain_lib.py` and `domain_rankmatched.py` are included in the public export, together with the sweep wrapper. The sweep still requires external aligned fields, date-freeze files and GIS rasters, which are listed with hashes in the data manifest; the archived passing-cell record validates the frozen run and is not a claim that the sweep runs without those inputs.

Road-modulated Hodge operator. The E/F/G comparison asks whether the spatial arrangement of the road field, rather than its marginal statistics, changes a fitted response. On the grid 2-complex let B_2 be the edge-to-plaquette boundary operator and

$$L_1 = D^\top D + B_2 B_2^\top, \qquad \tilde{L}_1 = \frac{2L_1}{\lambda_{\max}(L_1)} - I, \tag{S17}$$

so that $\tilde{L}_1$ has spectrum in $[-1, 1]$. The frozen road medium is $W_R = \text{diag}(w_e)$ on interior faces, with w_e the projected road length normalized to unit mean on its positive support with floor 10^{-3} (`g_ten`, the construction used for U_R). It is not refitted. The operator family is

$$C_\theta(W_R) = P_0 \left[\sum_{k=0}^{K} a_k \frac{W_R T_k(\tilde{L}_1) + T_k(\tilde{L}_1) W_R}{2} \right] D^\dagger, \tag{S18}$$

where T_k are Chebyshev polynomials and the symmetrized product removes the ordering ambiguity of the two non-commuting factors. The response is linear in $\theta = (a_0, \ldots, a_K)$; K is selected from

$\{0, 1, 2, 3, 4, 6\}$ and κ from the pre-registered relative ridge grid, both on VAL only. TEST is read once for the selected model. The construction satisfies $DC_\theta = 0$ for every medium; for a uniform medium $C_\theta(cI) = 0$ exactly, so heterogeneous spatial registration is necessary for a nonzero response. The selected order reaches the grid boundary $K = 6$ in all three cities. The search therefore did not settle below the boundary, but this does not establish that a wider family would improve TEST performance; the grid was not widened after TEST.

The comparison uses the FACE-departure target,

$$R^2_{\mathrm{D,FACE}} = 1 - \frac{\sum_{t\in\mathrm{TEST}} \|\delta F^\circ_t - \delta \widehat{F}^\circ_t\|^2}{\sum_{t\in\mathrm{TEST}} \|\delta F^\circ_t\|^2}, \tag{S19}$$

with δ the TRAIN phase departure. This is the D/FACE target. It differs from the P/FACE score in Supplementary Table S2 in the target (departure versus raw) and from the ROAD-support D score in the scoring space (full FACE versus U_R); it is not a separate scoring convention.

Matched non-road media. Two surrogate families preserve selected properties of W_R while destroying its arrangement. The directionally matched Null-1 preserves the exact weight multiset and 26 lag, smoothness and cross-family statistics within pre-registered tolerances; accepted draws also satisfy small road correlation and reduced top-decile overlap. The coarse-matched Null-2 additionally preserves the road's $\sigma = 2$ cell-scale projection by permuting within 20 equal-count strata and requires the fine residual to be weakly correlated with the road field. Each candidate is fitted with the same operator family, split and selection rule; the primary protocol reselects (K, κ) per draw, and a fixed protocol refits at the road-selected parameters. Tail estimates use the Monte-Carlo floor $1/(N + 1)$ and are comparisons with the constructed family, not permutation p-values. Accepted draws are correlated by construction, so effective sample sizes are smaller than nominal counts. San Francisco admits no coarse-matched draw under the frozen criteria (0/199); its result supports directional specificity only.

Prior-and-residual control. The road operator supplies a prior $\widehat{F}^\circ_{\mathrm{prior}} = C_\theta(W_R)\delta x$, and an equal-capacity reduced-rank ridge learner is fitted to the residual $\delta F^\circ - \widehat{F}^\circ_{\mathrm{prior}}$. The learner uses the frozen rank grid $\{1, 2, 4, 8, 16, 32, 64\}$ and the frozen penalty grid, selected on VAL. The absolute increment is

$$\Delta R^2 = R^2_{\mathrm{joint}} - R^2_{\mathrm{learned}}, \qquad g = \frac{\mathrm{SSE}_{\mathrm{learned}} - \mathrm{SSE}_{\mathrm{joint}}}{\mathrm{SSE}_{\mathrm{learned}}}, \tag{S20}$$

where g is a relative residual-SSE reduction, not an R^2 increment. The control repeats the same procedure with 20 directionally matched priors; with 20 draws the tail floor is 1/21. Porto is also evaluated in four contiguous date blocks separated by a five-date embargo, with inner 75/25 TRAIN/VAL selection. These are blocked internal checks, not an external test.

Note S5 Baseline and comparison protocols

Graph WaveNet. The completed adaptation [5] estimates the contemporaneous source-to-routing map on the frozen date splits. The implemented `cc_lib.anomaly` mask, rather than its misleading "every row" docstring, defines a P/FACE score: TRAIN inputs and targets are phase-centred, whereas VAL/TEST inputs and targets remain raw. Predictions are projected with P_0, so zero divergence is imposed rather than learned.

The archived loop constructs the network before seeding the training routine, leaving weight initialization uncontrolled. The sealed protocol declares one integer before construction, seeds `random`, NumPy and PyTorch, and reruns the full four-configuration grid with VAL selection repeated inside every seed. Porto and Chengdu use five seeds; San Francisco uses four seeds common to all five folds. The archived cells remain identifiable as single draws. The final duplicate Porto fold-0 per-date record is authoritative. Graphormer's separate 60-epoch reduced-budget run is excluded from the matched comparison and does not bound its model class.

Supplementary Table S2 uses the archived single draws for all three cities and pools each city's per-date error numerators and denominators. Main Table 1 reports the separate D/FACE modern comparison and does not rank these P/FACE values. For San Francisco this gives 0.2540, rather than the arithmetic mean of five fold R^2 values, and the archived per-date records make the cell directly comparable with the pooled UrbanFlow score. Sealed seed means are 0.548 in Porto (range 0.515–0.588) and 0.315 in Chengdu (0.260–0.350); the sealed San Francisco summary is a fold mean (0.221, range 0.096–0.279) whose per-date errors were not archived, so it cannot be pooled and is not used for ranking. The archived Porto paired-date contrast, Graph WaveNet minus UrbanFlow, is -0.1102 (95% interval $[-0.1762, -0.0268]$, Holm-adjusted $p = 0.0088$); under the sealed protocol the same difference is -0.059 on the seed mean and -0.019 at the worst seed. UrbanFlow leads every realised seed in Porto and Chengdu by at least 0.019 and 0.090. These results concern this completed adaptation and P/FACE protocol, not deep learning generally or a D-frame result.

Gravity and OD overlap. The doubly constrained gravity model estimates OD flows from TRAIN margins and distance, selecting friction on VAL. Its native common-part-of-commuters score at 4×4-cell aggregation is an OD-overlap measure, not a routing R^2, and the two are not numerically comparable. Radiation and Deep Gravity were not run because the required population/opportunity or covariate inputs were not supplied; no numerical result is attributed to them [6, 7].

All-or-nothing loading. AON uses time-shortest paths with archived road length/speed costs, a 150 m endpoint-snap threshold, and no congestion equilibrium or capacity calibration. Gravity loading retains the top 150,000 zone pairs under the archived combined-demand rule. A TEST-fitted global amplitude is diagnostic rather than deployable; the fixed-amplitude arm uses $\alpha = 1$. Exact-TEST-OD loading has additional demand information and is not a source-only prediction. AON does not stand in for Wardrop user equilibrium [8], which was not run.

Chengdu OD is cut from the corpus occupancy flag as maximal occupied runs, with frozen cleaning breaks (fix gap > 15 min, implied speed $> 120\,\mathrm{km\,h^{-1}}$, displacement > 3000 m) and retention above 60 s and 100 m. It is not inferred from a displacement heuristic. The observed field counts crossings during $[t_k, t_k + 3600\,\mathrm{s})$, whereas the loaded field routes every trip whose origin lies in window k. Moreover, Chengdu's observed field is marched from every fix while its OD uses occupied runs only; these overlapping but different activity sets are part of the comparison protocol.

Table S2: Archived P/FACE inference and network-loading diagnostics. This is the original comparison retained separately from the D/FACE modern table; the two scoring frames are not ranked together.

Model	Metric	Porto	San Francisco	Chengdu
Source-to-routing inference				
UrbanFlow	$R^2_{\mathrm{P,FACE}}$	0.607	0.528	0.440
Full-face ridge	$R^2_{\mathrm{P,FACE}}$	0.701	0.611	0.572
POD regression	$R^2_{\mathrm{P,FACE}}$	0.701	0.637	0.575
Graph WaveNet	$R^2_{\mathrm{P,FACE}}$	0.497	0.254	0.275
OD estimation				
Gravity	CPC	0.525	0.464	0.555
Network-loading diagnostics				
Gravity + AON	R^2_{FACE} [a]	0.136	0.270	< 0.001
TEST OD + AON	R^2_{FACE} [a]	0.172	0.336	0.002
TEST OD + AON	R^2_{FACE} [b]	0.126	-0.533	-0.565

Inference rows share sources, splits and targets; TEST errors are pooled from per-date numerators and denominators. Gravity uses TRAIN OD margins; AON denotes all-or-nothing assignment. [a]Amplitude fitted on TEST. [b]Fixed amplitude, $\alpha = 1$. Graph WaveNet reports archived single draws; sealed-seed robustness and the archived Porto paired contrast are in this Note. Loading rows diagnose the implemented pipelines; Chengdu's loading is margin-dominated.

Supplementary Results

Notes S6–S11 report held-out coupling, daily scores, coherent response, anisotropy, structural tests, the D/FACE modern benchmark and the structured-identification negative result. The three-city P-frame physical reconstruction is a qualitative complement to the complete held-out evaluation.

Note S6 Held-out coupling and baseline results

Table S3: Three-city P/D/CLIM scores. P scores the raw target; D scores TRAIN-centred departures; CLIM is the source-free P/ROAD phase clock. Cov. is raw routing energy captured by roads. Errors pool held-out observations within each city.

			P				D		
City	Cov.	CLIM	UrbanFlow		Ridge	POD	UrbanFlow		Ridge
		ROAD	ROAD	FACE	FACE	FACE	ROAD	FACE	FACE
Porto	0.904	0.472	0.682	0.607	0.701	0.701	0.468	0.411	0.504
San Francisco	0.886	0.727	0.597	0.528	0.611	0.637	0.240	0.209	0.262
Chengdu	0.731	0.931	0.606	0.440	0.572	0.575	0.120	0.085	0.126

The three focal cities replicate a procedure, not a single operator transferred between cities. Porto has the strongest departure coupling. San Francisco retains positive D scores despite a raw climatology stronger than raw operator reconstruction. Chengdu separates a highly recurrent raw field from weak departure prediction: P/ROAD is 0.606, CLIM is 0.931 and D/ROAD is 0.120. Adding the D departure to the TRAIN climatology gives a full-field gain $\Delta R_I = (1 - R_{\mathrm{CLIM}})R_D$ of approximately 0.25, 0.07 and 0.008 in Porto, San Francisco and Chengdu. The evidence is therefore hierarchical rather than three equivalent validations: Porto shows a clear conditional relation, San Francisco a weaker but consistent one, and Chengdu a highly recurrent field in which

source information adds little beyond the daily pattern.

Paired-date bootstrap intervals for D/ROAD are 0.355–0.528 in Porto, 0.188–0.288 in San Francisco and 0.107–0.142 in Chengdu (10,000 resamples, dates resampled within folds). The source–routing pairing null permutes complete date blocks within TRAIN, VAL and TEST separately and refits the operator in each of 199 draws; null medians are approximately zero, and no Porto or San Francisco draw reaches the observed score (empirical $p = 1/200$). Chengdu has only three held-out dates, so its null and interval are descriptive rather than significance tests. Figure S2 shows the date-resolved values.

Full-face ridge and TRAIN-only POD equal or outperform the road restriction on FACE targets. D/FACE ridge scores are 0.504, 0.262 and 0.126, against road-restricted 0.411, 0.209 and 0.085. Roads therefore supply independently specified, interpretable support rather than the highest linear reconstruction accuracy. Interior-face counts in Porto, San Francisco and Chengdu are 1,984, 5,512 and 7,080; D-frame selected POD ranks are 958, 16 and 64. Table S3 consistently pools errors rather than averaging fold scores.

Three-city P-frame physical reconstruction. Figure S6 visualizes observed and predicted road-supported routing fields using the frozen P estimator with raw TEST inputs and targets. No phase climatology is added and no D departure is substituted. Porto and San Francisco preserve their archived `picked_i/picked_phase` snapshots. Chengdu selects the TEST date whose daily P error is closest to the median daily P error: `picked_i=2`, `picked_phase=6` (12:00 in the 06–23 h window), `picked_date=16304`. Its pinned-basis export is checked against the frozen R_{P}. These snapshots are qualitative complements to the complete pooled scores, not additional independent experiments.

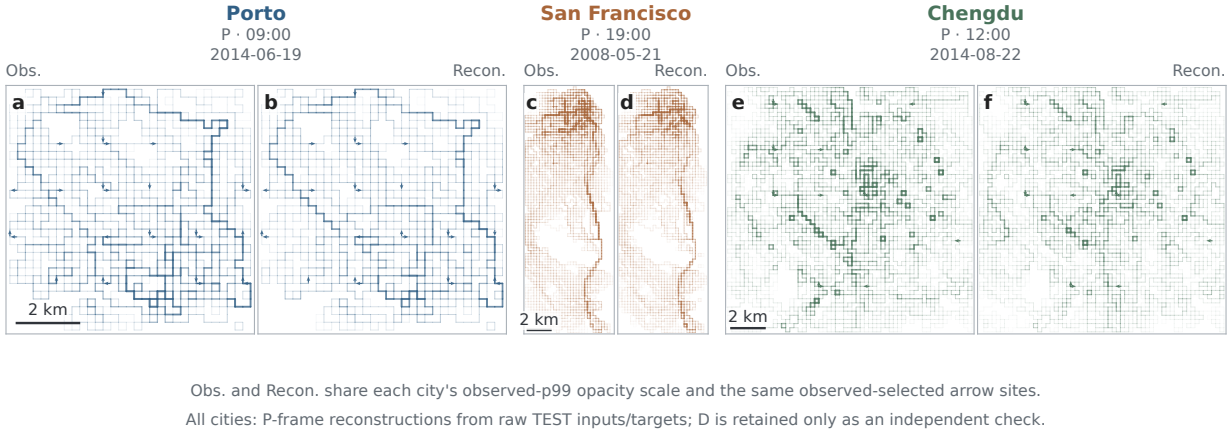


Figure S6: Three-city held-out physical reconstruction in the P frame. Observed and reconstructed road-supported routing fields for Porto, San Francisco and Chengdu, using the frozen P estimator with raw TEST inputs and targets; no phase climatology is added. Porto and San Francisco preserve their archived picked indices and phases. Chengdu uses the TEST date with daily P error closest to the median daily P error, specifically `picked_i=2`, `picked_phase=6` (12:00), `picked_date=16304`, and passes the pinned-basis/frozen-R_{P} export check. Observed and predicted panels within a city use identical glyph sites and the observed field's 99th-percentile magnitude scale, with saturation above it; colour scales are city-specific. The maps are qualitative, whereas Table S3 uses every held-out observation.

Neural aggregation. Seed configuration variability, the archived Porto paired contrast and the sealed seed means are reported in Note S5. Table S3 and Supplementary Table S2 use archived pooled per-date errors, including a San Francisco Graph WaveNet value pooled from its archived per-date records; the sealed San Francisco fold mean is not used for ranking.

Gravity and loading diagnostics. Native gravity CPC values are 0.525, 0.464 and 0.555. TEST-scaled gravity loading gives 0.136, 0.270 and 8.3×10^{-5}; exact-TEST-OD AON gives 0.172, 0.336 and 0.002, with fixed-amplitude values 0.126, −0.533 and −0.565. These tasks use different inputs and are not source-only accuracy rankings.

Chengdu's OD table contains 1.64 M TEST trips, of which 1.28 M remain after snapping. Its strong OD overlap but near-zero loading score is therefore not a ranking of demand quality. The loaded field has cosine 0.04 with the observed field, and destination permutation leaves the loaded field almost unchanged (cosine 0.93–0.94), showing that this pipeline resolves a largely margin-driven pattern. Rebuilding the observed three TEST dates from all fixes reproduces the frozen field at pooled cosine 0.999999. Using occupied runs alone removes 16.3% of absolute flux, leaves 76% of energy, and gives pooled cosine 0.929 with the all-fix field. Thus the activity-set difference is real and bounded, but by itself is unlikely to account for the loading mismatch. All loading rows diagnose the implemented demand–registration–loading pipeline rather than attainable performance of traffic assignment as a paradigm.

Note S7 Joint-spectrum, daily-score and coherent-response results

The joint field spectra, daily P/D scores and coherent response answer different questions. No operator is refitted for the spectral displays. The coherent comparison uses the same per-date DFT support as the joint spectrum in each city.

Three-city joint spectra. At the eight-mode display cut in Figure S1, source energy coverage is 0.6343, 0.9656 and 1.0000 in Porto, San Francisco and Chengdu, and routing coverage is 0.7989, 0.9892 and 1.0000. Chengdu's complete coverage follows from its eight-TRAIN-date snapshot-rank ceiling and does not demonstrate greater intrinsic low dimensionality. The common logarithmic floor is $10^{-4.5}$; all displayed cells exceed it. Chengdu's time-independent component carries 0.668 of source energy and 0.811 of full-face routing energy. The weighted spectra account for raw-block energy to machine precision; this is a numerical conservation check, not held-out evidence.

Held-out daily P/D scores. Figure S2 resolves the pooled entries in Table S3 by TEST date. Porto has 51 TEST dates, San Francisco has five blocked folds and Chengdu has three TEST dates in the continuous-observation protocol. The daily spread is descriptive: it does not replace the common pooled-error convention and does not supply the missing paired-date interval for the primary D comparison. In particular, Chengdu's three dates are insufficient for a strong date-heterogeneity claim.

Phase-sensitive frequency response. Figure S3 compares the frozen P-selected operator with observed routing on TRAIN and TEST using the phase-insensitive magnitude. Because that magnitude is unchanged by a common phase rotation, main Fig. 3 reports the per-bin phase-sensitive R^2_m and signed alignment. In Porto, R^2_m falls from 0.892 at $m = 0$ to 0.198 at $m = 12$, while $|\gamma_m|$ remains 0.44 at the highest bin. San Francisco has negative R^2_m at four bins (minimum

-0.281), and Chengdu at one (bin 8, -0.027). Pooled signed alignment spans 0.46–0.96 in Porto, 0.10–0.93 in San Francisco and 0.35–0.82 in Chengdu, so high magnitude alignment can coexist with substantial phase and amplitude error. The full per-bin and per-date values are archived in `reports/review_response/results/phase_metrics.json`; the weighted sum of the per-bin R^2_m reproduces each pooled R_{P} to machine precision. These results replace the earlier interpretation of γ_m as evidence of a correct complex response.

Underpowered Chengdu full-FACE nonlinear exploration. Figure S7 reports an exploratory 18-phase A-to-D comparison on the complete FACE target. For each nonlinear arm, ΔR^2 is the mean-date full-FACE error of `linF` minus that of the arm, so positive values favour the nonlinear arm. The `quadF` contrast is -0.005854 with descriptive 95% interval $[-0.006105, -0.005661]$; the `krrF` contrast is $+0.000418$ with interval $[-0.000023, +0.000698]$, which crosses zero.

There are only three TEST dates, and D's all-phase linear design has 144 columns for 144 TRAIN rows and is therefore saturated.

The design is **underpowered**: these arms neither establish a nonlinear improvement nor support a claim of no nonlinearity.

Full-FACE scores must not be mixed with POD-internal scores. Although 9,999 paired-date bootstrap draws are recorded, three dates provide only 27 ordered date resamples, so the intervals are descriptive rather than confirmatory.

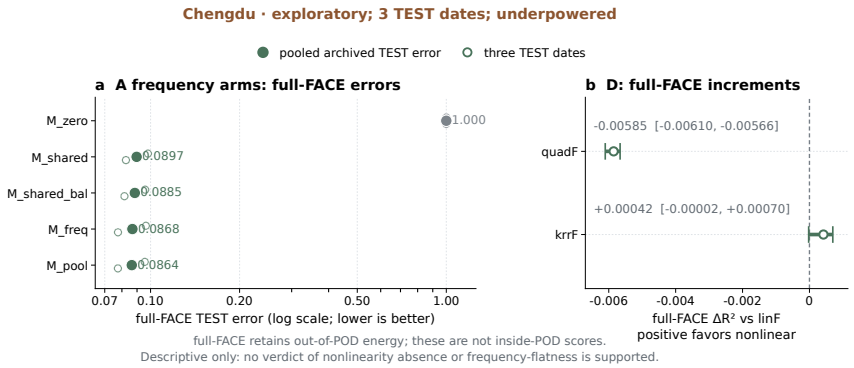


Figure S7: Underpowered exploratory Chengdu full-FACE nonlinear comparison. Full-FACE ΔR^2 relative to `linF` for `quadF` and `krrF` on the 18-phase A-to-D design, with descriptive 95% paired-date intervals. `quadF` gives -0.005854 $[-0.006105, -0.005661]$; `krrF` gives $+0.000418$ $[-0.000023, +0.000698]$. Only three TEST dates are available, and the 144-column linear design saturates the 144 TRAIN rows. The experiment is exploratory and underpowered; it does not demonstrate either the presence or absence of nonlinearity.

Note S8 Three-city anisotropy results

Median response anisotropy is 0.635, 0.651 and 0.705 in Porto, San Francisco and Chengdu, against structural-tensor medians 0.144, 0.057 and 0.280. The response exceeds structural anisotropy at 79.3% of Porto cells, 71–72% of San Francisco cells and 72.4% of Chengdu cells. These are properties of the fitted response on its identified source span, not of the instantaneous flow distribution. Figure S4 shows the full distributions behind these medians. Its zero-response atoms are distinct from low but nonzero anisotropy, and its rank-one atoms explain the vertical approach at $A = 1$.

Strong local anisotropy does not imply one city-wide preferred direction. Unweighted nematic coherence is 0.087 in Porto, 0.21–0.22 across San Francisco folds and 0.057 in Chengdu. Principal axes vary across space rather than collapsing onto one compass orientation.

Using the joint $A_C A_U$-weighted estimand in Eq. S12, the archived structural-axis similarity is 0.79, 0.79 and 0.76 in Porto, San Francisco and Chengdu. This is the single primary estimate. The spectrum-matched rotation null has minima 0.967, 0.902 and 0.803 and medians 0.973, 0.911 and 0.808; every null draw exceeds the archived estimate, so empirical $p = 1.0$ (Note S4). Axis agreement is therefore not distinguishable from the spectrum-matched null and provides no independent evidence of a road-aligned response. San Francisco's anisotropy map is reproducible across five folds, with mean pairwise spatial correlation 0.984 (range 0.974–0.991) and anisotropy-weighted nematic axis agreement 0.993. The invariance checks concern scalar anisotropy and covariant axial/tensor transformations as specified in Note S4; they do not imply an axis that remains numerically unchanged under a rigid physical rotation.

A separate amplitude control limits the road reading. On the Porto operator, the source-energy-weighted mean response anisotropy is 0.26, below the 0.32–0.33 range produced by four matched non-road media, and the cellwise Spearman correlation between response anisotropy and road coverage is -0.18 (999-draw cell permutation; one-sided $p = 1.00$ against positive association). The median values above describe the distribution of local anisotropy; they are not evidence that the anisotropy is specific to the road medium.

Note S9 Road specificity and matched-capacity structural results

Rank-matched subspace specificity. The structural test uses each city's full aligned observation pool, not the held-out P/D pool. Full-pool road captures are 0.900, 0.900 and 0.731 in Porto, San Francisco and Chengdu. Every one of the 999 random conservation-compatible subspaces leaves more routing energy uncaptured than the real road subspace (0/999 at least as good, empirical $p = 1/1000$). The excess $\Delta E = E_{\text{null}} - E_{\text{road}}$ ranges from 0.23 to 0.30 in Porto, 0.28 to 0.33 in San Francisco and 0.09 to 0.15 in Chengdu. This is a null distribution, not a confidence interval. The ambient $\ker D$ and rank are fixed while the random subspaces vary; the result concerns representation specificity and does not show that a source predicts the state within the road space.

Matched-capacity media. At $R = n_{\text{int}} = (1984, 5512, 7080)$ nominal pre-normalization block values, the constant-medium held-out R^2_{cond} values are 0.613, 0.743 and -0.030. Road-medium gains over that reference are 0.009, 0.010 and -0.001; mobility-inferred gains are 0.092, 0.085 and 0.036; and permuted-mobility-medium gains are -0.047, -0.052 and -0.016. The paired mobility-minus-road differences are 0.083 in Porto (95% interval 0.060–0.109), 0.075 in San Francisco (0.062–0.091) and 0.038 in Chengdu (0.035–0.040). Figure S5 shows all three media at this capacity.

The comparison matches capacity and estimator, not information: the road pattern is an external GIS raster, whereas the mobility-inferred pattern is estimated from TRAIN mobility. Both use the same nominal block capacity under the same normalization. Permuting the mobility medium's spatial arrangement preserves its values, partition and nominal block count and removes the gain. Chengdu uses one all-motion channel, a frozen per-fold basis and four held-out alternate-hour dates; its interval resamples only those dates. The design cannot separate the influence of date count, delivery phase and channel, and the four-date interval is only a descriptive summary, not the 14-date P/D experiment.

Together, the two tests show that the road-induced subspace is not generic under the stated rank-matched null, while the road-only medium does not match the mobility-inferred medium

under this estimator and information regime. They do not show that road geometry lacks physical meaning, that all alternative road structures have been tested, or that source-only closure has been achieved. The source files, real R grids and missing-module definition gap are listed in Notes S4 and S12.

Table S4: Rank-matched and matched-capacity structural controls. Full results from the former main Fig. 4d–e. The rank-matched test uses the full aligned observation pool; the matched-capacity test uses the date-centred conditional score.

Control	Porto	San Francisco	Chengdu
Road capture, full pool	0.900	0.900	0.731
Rank-matched nulls at least as good	0/999	0/999	0/999
Excess ΔE range	0.23–0.30	0.28–0.33	0.09–0.15
Road-medium gain at $R = n_{\text{int}}$	0.009	0.010	−0.001
Mobility-medium gain at $R = n_{\text{int}}$	0.092	0.085	0.036
Paired mobility − road [95%]	0.083 [0.060, 0.109]	0.075 [0.062, 0.091]	0.038 [0.035, 0.040]

Road-modulated operator in the FACE-departure frame. The road-medium test uses a different target from the rank-matched and matched-capacity tests above: the full divergence-free departure δF_t°, scored by Eq. S19. The frozen road operator reaches 0.0619 in Porto, 0.0649 in San Francisco and 0.0039 in Chengdu. These values are not comparable with the P/FACE entries of Supplementary Table S2 or the ROAD-support D scores; the unrestricted FACE ridge on the same rows reaches 0.5044, 0.2582 and 0.1258. A uniform medium gives a response at floating-point zero, and face permutation or a 90°/270° rotation of the medium collapses the held-out score to at most 0.005. The operator's selected spectral order is at the pre-registered grid boundary in all three cities.

Table S5: Road-medium contrast in the FACE-departure frame. Held-out R^2 for the road-modulated operator, directionally matched Null-1 and coarse-matched Null-2; "best" is the highest accepted draw. The unrestricted FACE ridge is shown for scale only. Dashes mark families that could not be constructed under the frozen criteria.

Score	Porto	San Francisco	Chengdu
Road operator	0.0619	0.0649	0.0039
Null-1 mean	0.0011	0.0040	0.0003
Null-1 best	0.0145	0.0273	0.0018
Null-2 mean	0.0046	–	0.0010
Null-2 best	0.0259	–	0.0019
FACE ridge (scale)	0.5044	0.2582	0.1258

The nominal draw counts are 499/199/199 for Null-1 and 99/–/49 for Null-2 in Porto/San Francisco/Chengdu; participation-ratio effective sizes are 148/93/175 and 5.6/–/12.4. The corresponding Monte-Carlo floors are 0.002/0.005/0.005 and 0.010/–/0.020.

A directionally matched non-road medium, preserving the exact weight multiset and 26 spatial statistics within the frozen tolerances, does not reproduce the road score: 0 of 499 accepted draws in Porto and 0 of 199 in San Francisco and Chengdu reach it. The best draws reach 23%, 42% and 47% of the road value. In Porto and Chengdu, coarse matching raises the surrogate mean

by factors of 4.1 and 3.6 over Null-1, respectively, but the road remains 2.4 and 2.1 times the best coarse draw. San Francisco has no accepted coarse-matched draw (0 of 199), so its evidence supports directional but not fine-scale specificity. The coarse families contain only about six and twelve effective independent configurations, and their tail estimates sit at the design floor; they bound the constructed family rather than calibrating a tail probability.

Road prior beyond a learned learner. Used as a prior under an equal-capacity residual ridge, the road operator raises held-out R^2 by 0.0116 in Porto, 0.0207 in San Francisco and 0.0023 in Chengdu. These are relative residual-SSE reductions of 2.27%, 2.65% and 0.26%, respectively. The 20 directionally matched priors change held-out R^2 by between -0.0013 and $+0.0003$. The road increment is positive on every Porto TEST date and in three of four blocked Porto folds; the fourth, highest-energy block is negative. The increment is largest when TRAIN is reduced, consistent with a prior that substitutes for data, and the Porto learner sits at the top of its rank grid, so the measured increment is a lower bound in a saturated regime. The prior-only score is still far below the learned model, and the learned correction has no reproducible response direction across TRAIN halves (median cosine 0.20–0.46). The evidence is for an informative structural prior, not an identified learned operator.

Independent static urban fields. A separate test asks whether other measured urban structure can replace the road medium. On the same Porto rows, operator family, (K, κ) grid and score, we fitted media from GHSL 2015 population density, GHSL 2015 built-up surface and Urban Atlas 2012 built land-use fractions (eight built classes; the five transport classes are excluded). Each cell field is mapped to faces as the mean of its two adjacent cells and normalized to unit mean with floor 10^{-3}. The additive joint medium is $0.5(W_{\mathrm{road}} + W_{\mathrm{nr}})$ after separate normalization; the multiplicative sensitivity is $\mathrm{normalize}(W_{\mathrm{road}} \cdot W_{\mathrm{nr}})$. The rasters were registered to the frozen row order by a single vertical flip determined from the static road layer, not from any mobility target.

Table S6 reports held-out D/FACE R^2 and paired TEST-date increments on the 51 TEST dates. The road medium reaches 0.0619; the three fields on their own lie between -0.0025 and $+0.0029$, and the additive joint media reach 0.0208–0.0380. Joint minus field is positive for all three fields ($+0.0200$, $+0.0384$, $+0.0233$), whereas joint minus road is negative (-0.0390, -0.0239, -0.0411). The complementarity is one-directional: the road medium adds to each field, but no field adds to the road medium. The multiplicative land-use composition does not clear the pre-registered 0.01 margin ($+0.0023$), so the conclusion is tied to the additive composition. All media remain weak in absolute terms. The result says that the measured fields do not substitute for the road medium; it does not say that road is the only possible medium.

Table S6: Independent static urban fields against the road medium (Porto). (a) Pooled TEST D/FACE R^2 in the frozen E2 family. (b) Paired TEST-date bootstrap increments, $\delta = 0.01$. Fields are external GIS layers with no mobility input; the joint medium is additive and the multiplicative composition is a sensitivity check. All values are from the frozen J3 output.

Quantity	Value	95% interval
(a) Held-out R^2		
Road	0.0619	–
Population	0.0029	–
Built-up surface	−0.0004	–
Built land use	−0.0025	–
Joint, population	0.0229	–
Joint, built-up	0.0380	–
Joint, land use	0.0208	–
Joint$_\times$, population	0.0152	–
Joint$_\times$, built-up	0.0225	–
Joint$_\times$, land use	−0.0002	–
(b) Paired increments		
Joint − population	+0.0200	[0.0135, 0.0231]
Joint − built-up	+0.0384	[0.0275, 0.0436]
Joint − land use	+0.0233	[0.0178, 0.0261]
Joint − road (population)	−0.0390	[−0.0439, −0.0293]
Joint − road (built-up)	−0.0239	[−0.0276, −0.0168]
Joint − road (land use)	−0.0411	[−0.0474, −0.0280]

Note S10 Modern D/FACE benchmark

This note reports the separate D/FACE comparison used in Results 1.5. It is not a re-ranking of the P/FACE scores in Supplementary Table S2: the target is the full divergence-free departure δF_t°, the source is δx_t, and every arm uses the same frozen splits, pooled ratio-of-sums score, VAL selection and single TEST pass. The two frames answer different questions and their numbers are never combined.

Protocol. The linear reference is the unrestricted FACE ridge (and POD, which solves the same full-rank problem) from the frozen three-city analysis. Four modern families were trained: a residual MLP with an explicit linear skip, a 2-D U-Net, a Fourier neural operator and GraphGPS. Each main arm used five seeds; the 300-epoch budget, VAL selection and early-stopping rule were fixed before any TEST row was scored. The scientific margin was $\delta = 0.01$ pooled R^2. Intervals resample TEST dates and recompute the pooled contrast. P/FACE values were retained as an auxiliary frame but never ranked against D/FACE.

Porto results. The linear reference reaches 0.5044. No neural arm reaches it. The best is the MLP with road channels at 0.4473, a gap of 0.0513 with a date-level interval excluding zero; the same MLP without road channels reaches 0.4213. The U-Net reaches 0.1827 without road channels and 0.1616 with them, the Fourier operator 0.0288 and 0.1097, and GraphGPS 0.1575 and 0.0193. The best neural configuration is the one closest to a linear map: a depth-0 MLP reaches 0.4916, and removing its nonlinear residual stack gains 0.0703. The ordering is therefore not a failure to train; it is a sample-size and target-dimensionality effect at the tested capacity.

Table S7: Modern D/FACE benchmark on Porto. Pooled TEST R^2 for the full divergence-free departure. Five seeds per neural arm; the linear reference is a single frozen fit. The VAL-reselected rows select the configuration on VAL alone; the linear contrast adds the frozen E2 road response and, in L2, two same-size matched null response blocks. The P/FACE values in Supplementary Table S2 are a different estimand and are not comparable.

Arm	D/FACE R^2	Seeds
FACE ridge / POD (linear reference)	0.5044	–
MLP, source only	0.4213	5
MLP, source + road	0.4473	5
U-Net, source only	0.1827	5
U-Net, source + road	0.1616	5
Fourier operator, source only	0.0288	5
Fourier operator, source + road	0.1097	5
GraphGPS, source only	0.1575	5
GraphGPS, source + road	0.0193	5
VAL-reselected		
MLP, source + road (cfg 4)	0.4404 (0.0208)	5
U-Net, source + road (cfg 1)	0.1589 (0.0052)	5
Fourier operator, source + road (cfg 3)	0.1414 (0.0153)	5
Linear road-structure contrast		
Ridge + E2 road block (L1)	0.5405	–
Ridge + E2 and two matched blocks (L2)	0.5343	–

Independent VAL reselection. The MLP, U-Net and Fourier grids were rerun with the configuration selected on VAL alone, and TEST was scored once. The ordering is unchanged: the best arm is the MLP at 0.4404 (s.d. 0.0208) against 0.5044 for the linear ridge, followed by the U-Net at 0.1589 (0.0052) and the Fourier operator at 0.1414 (0.0153). The selected configurations are MLP cfg 4, U-Net cfg 1 and Fourier cfg 3; the training budget, splits and score are unchanged. The seed base differs from Task H for configurations other than U-Net's, so these are independent replicate arms rather than a bitwise rerun.

Linear road-structure contrast. With the frozen E2 operator $s = C_{E2}^{\mathrm{phys}} x$, the feature block $[x, s]$ reaches 0.5405, an increment of +0.0361 over the 0.5044 ridge (95% date interval [0.0283, 0.0418]; positive on 51/51 TEST dates). Adding two same-size matched-null response blocks, $[x, s, s_{\mathrm{null1}}, s_{\mathrm{null2}}]$, raises VAL from 0.3422 to 0.3492 but lowers TEST from 0.5405 to 0.5343 (+0.0299 [0.0237, 0.0367] over the ridge). The increment is therefore not explained by the number of added features: the real road block transfers, the matched blocks do not. A null-only arm was not fitted, so the two-null contrast is the available control.

Replication and boundary. San Francisco and Chengdu reproduce the ordering but have three TEST dates each. The best neural arm reaches 0.2319 against 0.2582 for the linear reference in San Francisco, and 0.0125 against 0.1258 in Chengdu. These are sign-and-ordering replications, not significance claims. Road channels help the Fourier operator consistently, help the MLP on the pooled estimand but not on the unweighted date mean in Porto, and hurt GraphGPS and the Porto U-Net; the effect is architecture-dependent. The comparison concerns four architectures at one capacity, one sample size and one 300-epoch budget. It does not establish that neural models

cannot learn this map, and it does not re-run UrbanFlow. Exact projection, where applied, costs at most 0.0066 and is retained for admissibility.

Note S11 Structured operator identification: a negative result

This note reports the Porto identification experiment referenced in Discussion 2.2. It is deliberately a negative result about one estimator, one 14-atom dictionary and one city; it is not an XAI method and it does not revise the road-specific evidence from the matched-medium tests.

Estimand and models. The target is the FACE-departure field δF_t°, the source is δx_t, and the score is the pooled D/FACE R^2 used above. I0 is the learned-only full-face ridge. I1 adds a fixed five-atom E2 road prior selected on VAL. I2 fits a 14-atom structured dictionary jointly with the residual ridge; I4 is the same dictionary with a ridge rather than lasso penalty. I3 repeats I2 with four structure-matched surrogate media. Every arm uses the same splits, VAL rule, penalty grid and single TEST pass.

Table S8: Porto structured-operator identification. Pooled D/FACE TEST R^2; the I3 entry is the range over four qualified surrogate-media draws. The structured share is the TEST R^2 of the prior applied without the residual ridge.

Model	End-to-end R^2	Structured-only R^2
I0 learned-only ridge	0.5044	–
I1 fixed E2 prior	0.5139	–
I2 14-atom joint fit	0.5262	0.1393
I3 matched surrogate media	0.5249–0.5265	0.1325–0.1443

Why the gain is prediction-only. I2 improves the end-to-end score by 0.0218 over I0. Every surrogate medium reproduces essentially the same gain, and one surrogate lies above the real medium. The structured part alone explains 0.1393 of TEST variance, while the surrogates span 0.1325–0.1443; the real medium sits inside that range. The lasso path selects $\lambda = 0$ and all 14 atoms, so no compact support is identified. The standardized TRAIN Gram has condition number 9.2×10^{15} and mean absolute off-diagonal correlation 0.42; the coefficient vector is a minimum-norm representative of a nearly degenerate design, not a set of physical road parameters. The response is stable when the dictionary basis is fixed (half-TRAIN cosine 0.997), but this stability does not establish medium identity.

Post-hoc attribution control. Integrated gradients on the road-conditioned MLP are faithful for the source field: deleting the top cell costs about 11 points of VAL R^2, far beyond a matched random control. For the road block, deletion is much weaker (2.5 points at the top 32 entries) but still above its control. However, models trained on the real and spatially permuted road fields differ by 0.043 TEST R^2, while their attribution maps are indistinguishable in magnitude, shape and stability. Post-hoc attribution therefore cannot certify the road medium. The negative result is consistent with the matched-medium findings above: the estimator can use a structured prior without identifying which medium generated it.

Supplementary Reproducibility

Note S12 Artifact map and reproducibility

Table S9 separates frozen-score recomputation, figure rebuilding and incomplete raw/training exports. All paths are repository-relative. A file's presence does not imply that an end-to-end raw-trajectory rebuild is possible, and missing modules or per-run outputs are stated rather than replaced by inferred definitions.

Table S9: Artifact and reproducibility map. Paths are relative to the repository root. The status column distinguishes archived evidence from entry points runnable in the present export.

Result family	Source artifacts and code	Reproduction status or limitation
P/D/CLIM and linear controls	`results/frozen/crosscity/crosscity_results.csv`; `results/frozen/constitutive/obj_*.json`; `results/frozen/constitutive/d_uncertainty.json`; `experiments/constitutive_coupling/d_uncertainty.py`	Frozen per-fold objects and pooled scores, plus paired-date bootstrap intervals and source–routing pairing nulls. Chengdu's three held-out dates make its interval and null descriptive.
Daily scores, phase metrics and coherent response	`results/frozen/constitutive/obj_*.json`; `results/frozen/constitutive/chengdu_response18.npz`; `results/frozen/spectra/phase_metrics.json`; `experiments/frequency_response/phase_metrics.py`	Frozen date-level diagnostics, per-bin phase-sensitive R^2_m and signed alignment. The weighted per-bin scores reproduce each pooled R_{P} to machine precision.
Analysis footprint mask	`paper/merged_manuscript/supplement_sections/01_data.tex`	The archived footprint is $(N > 0)$.any(0) over all dates and is transductive. It is used only for display and analysis cells, not to construct D, U_R or K_R.
Three-city physical reconstruction	`results/frozen/constitutive/chengdu_snapshot18.npz`; `results/frozen/constitutive/chengdu_frozen_response_manifest.json`; `experiments/field_construction/chengdu_frozen_response.py`; `results/frozen/reconstruction/reconstruction.npz`	Chengdu uses the P estimator with raw TEST inputs and targets and passes pinned-basis and frozen-R_{P} checks. Porto/San Francisco full snapshots are not bundled; the compact display product is.
Chengdu nonlinear exploration	`configs/experiments/chengdu_nonlinear.json`; `experiments/baseline_comparison/chengdu_nonlinear_exploratory.py`; `results/frozen/baselines/chengdu_nonlinear/chengdu_nonlinear_exploratory.json`; `results/frozen/baselines/chengdu_nonlinear/validation.json`	The 18-phase full-FACE A-to-D experiment is exploratory with three TEST dates and 144 TRAIN rows. It supports neither nonlinear improvement nor absence of nonlinearity.
Road-modulated operator and matched media	`results/frozen/road_medium/`; `reports/roundE_graph_constitutive/outputs/`; `reports/roundF_constitutive_evidence/outputs/`; `reports/roundG_directional_medium_null/outputs/`	Aggregate E2/E3, F3 and G4 results are bundled in `results/frozen/road_medium/`. Full null-draw archives and construction scripts remain in the analysis-tree rounds named here and are not part of the public export.
Modern D/FACE benchmark (Task H)	`results/frozen/modern_benchmark/`; `reports/roundH_modern_baselines/outputs/agg/`	Aggregate summaries and paired-date CSVs are bundled. Model weights and full training runs are not. This D/FACE table is not comparable with the P/FACE Table S2.
Task J1–J4 rechecks	`reports/roundJ_claim_validation/outputs/`; `reports/roundJ_claim_validation/scripts/`	Frozen JSON/NPZ records for anisotropy controls, cross-regime splits, independent static fields and VAL-reselected arms. Analysis-tree records, not in the public export.
Structured identification (Task I)	`results/frozen/identification/`; `reports/roundI_interpretable_operator/outputs/runs/`	Porto identification, stability and attribution records are bundled as aggregate outputs. Training caches and model weights are not. The result is prediction-only, not a physical identification or a new XAI method.
Archived neural and sealed Graph WaveNet	`results/frozen/crosscity/crosscity_results.csv`; `configs/baselines/graph_wavenet.yaml`; `results/frozen/baselines/splits/`	Aggregate scores and frozen split records are bundled. Large predictions, Jacobians and complete sealed per-run JSON outputs are not included.
Gravity and network loading	`results/frozen/crosscity/crosscity_results.csv`; `configs/baselines/`	Aggregate scores are bundled; detailed loading JSON and the Chengdu collated record are not. Loading rows diagnose the implemented pipeline and are not source-only predictions.
Joint spectrum and anisotropy	`results/frozen/spectra/verified_joint.npz`; `results/frozen/spectra/joint_spectrum_audit.json`; `results/frozen/anisotropy/fo4_fig5_data.npz`; `results/frozen/anisotropy/chengdu_anisotropy.npz`; `results/frozen/anisotropy/anisotropy_null.json`; `results/frozen/anisotropy/gtheta_provenance.json`	Weighted Parseval display, source hashes, archived tensors and the 999-draw rotation null are bundled. Axis agreement is not independent validation.
Structural and matched-capacity tests	`results/frozen/anisotropy/partA_topology.json`; `results/frozen/anisotropy/partA_chengdu.json`; `results/frozen/anisotropy/partF_summary.json`; `results/frozen/matched_capacity/`; `archive/legacy/matched_capacity/`	Aggregate rank-matched and matched-capacity results are bundled. The original construction modules are archived and depend on external fields and GIS rasters.
Canonical figures	`figures/main/`; `figures/supplementary/`; `scripts/reproduce_figures/reproduce_all.py`	Four main figures and seven supplementary figures map to their generators. Figure rebuilding is distinct from recomputing the experiments.

Deep-baseline record. The archived scoring chain is `run_modern.py` → `bmlib.anomalies` → `cc_lib.anomaly`. Only TRAIN rows are phase-centred by the called implementation, leaving

VAL/TEST raw. Deduplicating Porto GWNET fold 0 by retaining the final authoritative per-date record gives an archived single draw of 0.4970176832 in Porto; pooling the archived San Francisco per-date numerators and denominators gives 0.2540477278. These are the values used in Supplementary Table S2, not the sealed seed means. Graphormer's 60-epoch budget and incomplete fold-level prediction-file export remain disclosed. Per-date error records exist for all scored archived folds, but zero-divergence residuals follow from projection and are not learned conservation.

Release scope and unresolved checks. The public export separates figure rebuilding from frozen products, core-score recomputation and end-to-end raw-trajectory reconstruction. It contains no individual-level trajectory records. Large prediction files, unavailable original training artifacts and incomplete sealed per-run outputs are stated rather than represented as included. The matched-capacity construction modules are included, but the sweep remains dependent on external aligned fields, date-freeze files and GIS rasters listed in the data manifest. Raw-data redistribution requirements, the OSM snapshot/version and per-artifact licence notices remain release checks (Note S1). Public repository location: https://github.com/Wangwangw233/paper-urbanflow.

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